

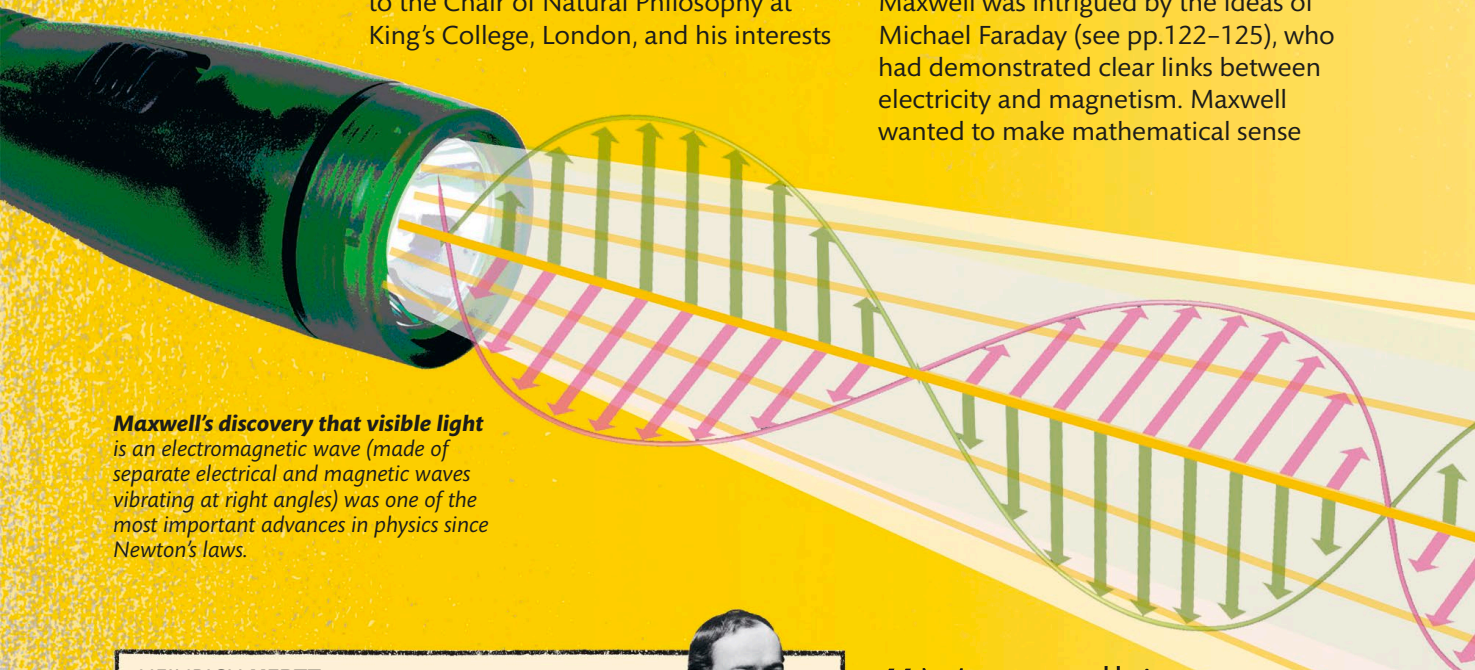
remain stable without breaking up. Maxwell spent 2 years working on the problem before showing that the rings could not be solid or fluid, but must be made of innumerable small particles. He won the prize in 1859. His analysis was ultimately proven in 1980, when NASA's Voyager 2 reached Saturn.

In 1860, Maxwell was appointed to the Chair of Natural Philosophy at King's College, London, and his interests

broadened further. He greatly advanced the kinetic theory of gases, showing the mathematical relationship between the temperature of a gas and the different speeds of its molecules. However, it is for his work on electromagnetism that he is most renowned.

A grand unified theory

Maxwell was intrigued by the ideas of Michael Faraday (see pp.122–125), who had demonstrated clear links between electricity and magnetism. Maxwell wanted to make mathematical sense



Maxwell's discovery that visible light is an electromagnetic wave (made of separate electrical and magnetic waves vibrating at right angles) was one of the most important advances in physics since Newton's laws.

HEINRICH HERTZ

German physicist Heinrich Hertz discovered radio waves and was the first to prove electromagnetic theory by experiment. The unit of frequency is named after him: the hertz.



Hertz (1857–1894) was a gifted young man who worked in a physics laboratory in Berlin. He was invited to enter a competition to verify James Clerk Maxwell's theory, but did not think he was capable. Years later, in 1886, he noticed that electric sparks caused a regular vibration, resulting from fluctuating electric charges. He realized that, if Maxwell was right, these would radiate electromagnetic waves. He designed the necessary apparatus and produced radio waves. In fact, he had made the first radio transmitter.

“Maxwell is the physicist's physicist. He is **the unsung hero of British science.**”

Stephen Hawking, 2010